Before the
United States Copyright Office, Library of Congress
Washington, D.C.

In the Matter of	)	
Section 1201 Study:	)	Docket No. 2015-8
Request for Additional Comments	)	

Additional Comments of Auto Care Association

Auto Care Association ("Auto Care") submits these additional comments in response to the September 27, 2016 Notice of Inquiry ("NOI").[1] Auto Care is a national trade organization of 3,000 members representing more than 150,000 independent businesses that manufacture, distribute, and sell motor vehicle parts, accessories, tools, equipment, materials, and supplies, and perform vehicle service and repair. As Auto Care explained in its Comments in this proceeding:

- The independent aftermarket industry employs more than 4,300,000 people in the United States.
- Nearly $341 billion dollars was spent on motor vehicle care in the United States in 2015, or more than 1.9 percent of the U.S. gross domestic product.
- Following expiration of a new car warranty, over 70 percent of car owners who patronize auto repair shops rely on independent repair shops over new car dealers.
- Approximately 20 percent of American consumers buy automotive parts and products to maintain, repair, and customize their own vehicles.

Original equipment manufacturers ("OEMs") increasingly use firmware and software to control the function and operation of motor vehicle parts.[2] In 2000, a typical SUV had fewer than ten (10) modules with embedded software controllers. By the 2015 model year, that same vehicle model incorporated 70 or more software units. Exhibit 1 illustrates more than 45 functions in a typical modern automobile that are controlled by

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[1] Auto Care previously submitted comments to the Copyright Office and the Librarian of Congress on these subjects, and participated in the Copyright Office Roundtables on May 18-20, 2016. *See* Comments of Auto Care Association, *In the Matter of Section 1201 Study: Notice and Request for Public Comment*, Dkt. No. 2015-8 (March 3, 2016), available online at https://www.regulations.gov/document?D=COLC-2015-0012-0036; Comments of Auto Care Association, *In the Matter of Software-Enabled Consumer Products Study: Notice of Inquiry and Request for Public Comment*, Dkt. No. 2015-16 (Feb. 16, 2016), available online at http://www.regulations.gov/#!documentDetail;D=COLC-2015-0011-0011.

[2] *See* Robert N. Charette, "This Car Runs on Code," IEEE Spectrum (Feb. 11, 2009), http://spectrum.ieee.org/transportation/systems/this-car-runs-on-code (last visited Oct. 20, 2016); National Instruments, "ECU Designing and Testing using National Instruments Products," (Nov. 7, 2009) *http://www.ni.com/white-paper/3312/en/* (last visited Oct. 20, 2016).

electronic modules with embedded software.[3] Software modules also control "telematics" that restrict wireless access to vehicle diagnostic information to only the OEM-authorized servicers.

Vehicle repair fills a critical need for consumers and business. According to the U.S. Department of Transportation, in 2014 more than 260,350,000 automobiles,[4] and more than 135,000,000 trucks and truck-tractors,[5] were registered in the United States. The need to repair, customize, or replace the software in these motor vehicles has attained greater importance as American consumers and businesses keep their cars and trucks in service longer. On average, new car buyers in 2015 will keep their vehicles for 6.5 years, and used car buyers keep those cars on the road for 5.3 more years – a total of more than 11-1/2 years, and more than 4 years longer than the average in 2006.[6] As cars advance in age, so does the need for consumers and businesses to repair and replace vehicle parts.

Vehicle repair also contributes to a cleaner environment. Repair and customization improves vehicle performance and efficiency, increases gas mileage and reduces energy consumption, and lowers emissions. Repair of motor vehicles using repaired, reconditioned, and rebuilt automotive parts typically re-uses 88 percent of the raw material from the original parts, and rebuilding engines saves 50 percent of the energy required to produce a new engine.

Consumers and businesses prefer to obtain parts and service from independent businesses rather than dealerships—benefitting from reduced costs, greater convenience, and higher consumer satisfaction.[7] Parts and service from independent manufacturers and repair shops cost less than OEM parts and service, and competition from these independents constrains supracompetitive pricing by OEMs. Independent repair businesses have the equipment, knowledge, and experience necessary to address the consumer's problem accurately and in compliance with laws and regulations pertinent to motor vehicles.[8]

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[3] Electronic control units in motor vehicles also may include databases or compilations, such as of numeric parameters used to control system performance. References to "software" in this comment should be read to include any such databases or compilations to the extent they are contended to be copyright-protected.

[4] U.S. Department of Transportation, Federal Highway Administration, Highway Statistics 2014, http://www.fhwa.dot.gov/policyinformation/statistics/2014/mv1.cfm

[5] *Id.*, http://www.fhwa.dot.gov/policyinformation/statistics/2014/mv9.cfm

[6] News Release, IHS, "Average Age of Light Vehicles in the U.S. Rises Slightly in 2015 to 11.5 years," July 29, 2015, http://press.ihs.com/press-release/automotive/average-age-light-vehicles-us-rises-slightly-2015-115-years-ihs-reports; Phil LeBeau, "Americans holding onto their cars longer than ever," CNBC (July 29, 2015) http://www.cnbc.com/2015/07/28/americans-holding-onto-their-cars-longer-than-ever.html

[7] "Independent vs. dealer shops for car repair," Consumer Reports, Jan. 22, 2015, http://www.consumerreports.org/cro/magazine/2015/03/best-places-to-get-your-car-repaired/index.htm

[8] In that regard, it is particularly regrettable that the triennial ruling denied the exemption to businesses, acting "on behalf of" their customers, that have the experience and wherewithal to both circumvent the technological protection measure and effect repairs compliant with

Consumers and businesses have the right to repair their motor vehicles at their preferred repair shops, and independent repair shops have the right to provide consumers with necessary parts and service. Cleaning, repairing, and refurbishing parts, and replacing worn or broken parts for automobiles has long been held permissible repair under patent law[9] and trademark law[10]—without imposing any restrictions on whether those repairs or refurbishments are performed by the owner of the article or someone acting on their behalf. The analogous right to repair a copy of a copyrighted work, consistent with the law of exhaustion and Section 109, cannot be doubted and should not be denied to either the owners or lessees of a copy or someone acting on their behalf. Especially, these rights should not be denied with respect to embedded software that controls functional operations of physical machines.

Yet, as the former Register observed in testimony last year, many consumers justifiably are concerned that copyright law suddenly interferes with their reasonable and customary rights to freely repair and customize the products they buy.[11] Auto manufacturers have deployed technological measures over their electronic control modules not to protect copyrights, but ultimately to thwart competition—to prevent independent servicers and consumers from making repairs and routine maintenance. Some manufacturers design their technological measures to preclude use of non-OEM replacement parts, such as by tying the copy of the embedded software to the car's unique Vehicle Identification Number, or by shutting down operations where the vehicle does not use "authorized" replacement parts.[12] In two cases, federal district courts have dismissed automaker complaints under the DMCA—in one case, where the technological measure protected access to the electronic control module itself, and not to the embedded

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environmental and other regulations. Notwithstanding, Section 1201 grants no authority to limit exemptions based on considerations other than the protection of copyrighted works.

[9] *Aro Mf'g Co. v. Convertible Top Replacement Co.*, 365 U.S. 336 (1961) (replacing fabric convertible tops); *Dana Corp. v. American Precision Co.,* 827 F.2d 755 (Fed. Cir. 1987) (rebuilding automobile clutches using new parts and used parts from many disassembled worn clutches held permissible repair).

[10] *Champion Spark Plug Co. v. Sanders*, 331 U.S. 125 (1947) (use of OEM trademark to accurately identify repaired and reconditioned auto parts is permissible under Lanham Act).

[11] *See* "The Register's Perspective on Copyright Review," Statement of U.S. Register of Copyrights Maria A. Pallante before the Committee on the Judiciary, U.S. House of Representatives at 23-34 (Apr. 29, 2015), available online at http://copyright.gov/laws/testimonies/042915-testimony-pallante.pdf.

[12] In the context of replacement parts for computer printers, s*ee Lexmark v. Static Control*, 387 F.3d 522, 530 (6th Cir.2004) (technological protection measure on printer cartridge chip prevented use of "unauthorized" toner cartridges); Alex Hern, "HP 'timebomb' prevents inkjet printers from using unofficial cartridges," The Guardian (Sept. 20, 2016). https://www.theguardian.com/technology/2016/sep/20/hp-inkjet-printers-unofficial-cartridges-software-update (last visited Oct. 20, 2016).

software;[13] and in another case, where technological measures were circumvented solely for the purpose of accessing non-copyrightable data and operating parameters in the embedded software.[14]

The focus of Section 1201 is on the protection of rights arising under copyright. Yet, these technological measures on their face protect business models, not rights granted under Section 106. Auto Care submits that these measures misuse Section 1201 just as the makers of garage door openers[15] and printer cartridges[16] did, by invoking laws intended to protect copyrighted works for the purpose of locking out competition for non-copyrightable parts and services. As the concurrence in the *Static Control* case warned:

> [M]anufacturers could potentially create monopolies for replacement parts simply by using similar, but more creative, lock-out codes. Automobile manufacturers, for example, could control the entire market of replacement parts for their vehicles by including lock-out chips. Congress did not intend to allow the DMCA to be used offensively in this manner, but rather only sought to reach those who circumvented protective measures 'for the purpose' of pirating works protected by the copyright statute.[17]

Auto Care urged the Copyright Office in its prior comments in this proceeding to reinforce the metes and bounds of Section 1201, and affirm that Section 1201 does not apply where the measure protects either the nonexpressive or noncopyrightable aspects of embedded software or the markets for physical objects controlled by the software.

Auto Care welcomes the invitation of the Copyright Office to provide additional comments and input on the solutions needed to secure the right to repair and to realign the DMCA with its intended purpose. Auto Care responds below to Subjects of Inquiry sections 1(c), 2, and 3.

I. Response to 1(C): A Permanent Exemption is Warranted for Circumvention of Access TPMs for Purposes of Analysis, Customization, Repair, and Replacement of Control Software in Motor Vehicles.

A. <u>Scope of the Proposed Permanent Exemption</u>

Auto Care would welcome the adoption of a permanent exemption for circumvention of software that controls the function and operation of motor vehicle products. Such an exemption should cover all activities necessary for the repair or customization of a motor vehicle, including:

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13 *General Motors LLC v. Dorman Products, Inc.*, No. 15-12917 (E.D. Mich. Sept. 30, 2016).

14 *Ford Motor Company v. AUTEL US Inc.*, No. 14-13760, Opinion and Order, (E.D. Mich. Sept. 30, 2015).

15 *Chamberlain Group v. Skylink Tech., Inc.*, 381 F.3d 1178 (Fed. Cir. 2004).

16 *Lexmark Int'l Inc., v. Static Control Components, supra* n.7.

17 *Id.*, 387 F.3d at 552 (Merritt, J., concurring).

- Ø accessing the software;
- Ø making interim copies of the software;
- Ø analyzing the software code;
- Ø altering the software code for the purposes of repair or customization, or developing a noninfringing software program to be used in lieu of all or part of the OEM-supplied software;
- Ø making a replacement copy of the software with any such alterations; and,
- Ø embedding that replacement copy in either the original part or a refurbished or replacement part.

Each of these rights must be clearly accommodated in the exemption so that every business and individual will have a full and meaningful right to repair.

The exemption also must encompass all steps necessary to analyze the embedded program. Diagnosis of the problem to be repaired and maintenance and repair of the motor vehicle all are essential to rectifying the performance and operational issues of concern to vehicle owners or lessees. However, the first step to solving consumers' needs is understanding what the embedded program is and how it works. Without that initial analysis, aftermarket businesses and consumers cannot correct, adjust, or adapt the software or develop an alternative solution.[18]

B. The Exemption Must Extend to Service Businesses that Act On Behalf Of <u>a Vehicle Owner or Lessee</u>.

This exemption cannot be limited to just an individual consumer or business who owns or leases a motor vehicle. An exemption, once justified, must be available equally to all members of the class. To achieve that end, all persons entitled to the exemption must be permitted to retain the services of third parties acting on their behalf.

As noted above, and in Auto Care's initial Comments in this proceeding, third party businesses have every right to provide vehicle owners with repair, maintenance, and customization services under patent law. Other federal and state laws protect consumers' right to obtain independent repair services, such as laws limiting OEM attempts to void equipment warranties.[19] There is no rationale by which copyright law should restrict such services where software essentially performs the mechanical functions of hardware, often substituting functions once performed by hardware alone.

Given the complexity of the electronic control units, specialized tools and knowledge often are required for customization and repair. Moreover, knowledge of cryptographic techniques and coding is necessary to circumvent technological locks. Servicers possess in-depth specialized knowledge of vehicle mechanics that benefits consumers. These servicers possess the

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18 To the extent that making a copy of the program facilitates that analysis, such intermediate copying should be deemed noninfringing fair use. *See, Sega Enterprises Ltd. v. Accolade, Inc.*, 977 F.2d 1510 (9th Cir. 1992).

19 *See, e.g.,* Magnuson-Moss Warranty Act, 15 U.S.C. § 2301 *et seq.*

tools and skill to efficiently and effectively diagnose and address vehicle issues that many consumers cannot or prefer not to tackle themselves. Servicers understand how particular vehicle electronic control systems interact with other software-controlled systems, and so can anticipate and adjust the repair to one system to correctly interoperate with the vehicle's entire software ecosystem. Many consumers today have even the most basic maintenance and repair services performed by a third party servicer. Thus, the only practical way to exercise this exemption is to allow repair services to also provide more complex software-controlled maintenance, repair, and customization as well. Without an "on behalf of" right, the exemption can be easily nullified by interposing a proprietary physical interface or by increasing the complexity of the program code of the technological measure beyond the skills of all but a few of the hundreds of millions of consumers who otherwise should benefit from the exemption. Where class members lack the means to benefit from the exemption, Auto Care submits the Copyright Office has a duty to make the exemption meaningful by extending the exemption to servicers acting on behalf of the vehicle owner or lessee.

Auto Care therefore continues to support an exemption along the lines of the petitioners' proposal:

> *The exempted class allows circumvention of a TPM protecting a computer program that controls the functioning of a motorized land vehicle, including personal automobiles, commercial motor vehicles, and agricultural machinery, for purposes of lawful analysis, diagnosis and repair, or aftermarket personalization, modification, or other improvement, including programs that modify the code or data stored in such a vehicle and including compilations of data used in controlling or analyzing the functioning of such a vehicle. The exemption further allows circumvention when undertaken by or on behalf of the lawful owner or lessee of the vehicle, or of a computer to which the computer program or data compilation relates.*

C. A Permanent Exemption Alone is Not Sufficient; an Exemption is Needed for the Creation and Distribution of Circumvention Tools.

An exemption that extends only to the act of circumvention will not be widely available to all members of the class. Extending the exemption to servicers who act on behalf of class members partially ameliorates this disparity of access, but does not fully resolve it. To provide effective service to vehicle owners, servicers must be able to circumvent technological measures and repair software in dozens, if not hundreds, of vehicle makes and models. The complexity and cost involved may prove prohibitive for servicers, whose employees typically are expert in auto mechanics rather than cryptography or coding. The resulting expense of individualized circumvention and software analysis and repair would make the exemption impractical, thus driving consumers back to the OEM dealership and effectively nullifying the purpose and benefit of the exemption.

The rational solution to this dilemma is to permit companies with expertise in software development to develop and make circumvention and repair solutions available to servicers and consumers. This provides obvious benefits to the market and the public. Vehicle repair software will be of consistently high quality and reliability. And the costs of circumvention and software repair can be spread among the many thousands of servicers that acquire the solution, rather than be shouldered by each individual consumer. Moreover, a right to distribute these software solutions creates no undue risks to copyright owners. Because the circumvention tools will be specialized for particular types of automobile software, the tools themselves would only enable competition for repair of embedded software without creating general risks of circumvention or infringement for non-specialized computer software or expressive copyrightable works.

Auto Care therefore respectfully submits that, if legislation is contemplated or required to secure a meaningful right of vehicle repair, such legislation must encompass the right of servicers to engage in repair on behalf of the class of vehicle owners and lessees; the right to circumvent for purposes of analysis as well as diagnosis, maintenance, and repair; *and* the right to create circumvention tools and replacement software that can be made available to such servicers and consumers.

D. Existing Legal Constructs Do Not Fully Address the Needs of the Exempt Class.

As Auto Care wrote in its initial Comments in this proceeding, neither Section 117 nor Section 1201(f) adequately and clearly addresses the right of consumers to repair vehicles with embedded software. Processes necessary to perform vehicle repairs are not limited to the making of archival copies or copies in temporary memory, and the circumvention at issue may not be solely for the purpose of facilitating interoperability with independently created programs. None of these exceptions is sufficient to enable vehicle repair, customization, and maintenance. Similarly, while these activities would constitute fair use under Section 107, as a practical matter a finding of fair use arises only as a defense to a claim of infringement. The expense of litigation, and the threat of statutory damages, would inevitably chill competition in the vehicle repair aftermarket from services and software developers that otherwise indisputably would be entitled to engage in repair under patent and trademark law.

Likewise, with respect to Inquiry Section 2, while the Section 1201(j) exemption for security research, the 2015 exemption for good-faith security testing, and the Section 1201(g) exception for encryption research remain valuable for other purposes, none of these exemptions addresses the needs of vehicle repair.

Finally, the Massachusetts "Right to Repair" law and the voluntary Memorandum of Understanding with the vehicle manufacturers which extended the requirements to the entire country do not adequately address the consumer issues created by embedded software. The right to repair laws and the national agreement only mandate that car companies make available to independents the same repair information, tools, and software that they make available to their dealers. These laws do not cover the need by repair shops to use or copy software for the installation of aftermarket parts that must work with other vehicle systems with embedded software, or the needs of companies that could develop such parts. Ultimately, neither state laws nor limited voluntary agreements eliminate the impediments erected by the overbreadth of Section 1201. Only federal legislation or regulation can ameliorate those flaws.

II. Anti-Trafficking Provisions Must Not Apply to Tools that Enable Lawful Vehicle Repair.

As described in Auto Care's initial Comments, and as noted above, Auto Care members' experiences show that, in the case of embedded software, both the act of circumvention and the ability to make effective noninfringing uses can be technologically beyond the reach of the ordinary consumer. In such circumstances, copyright policy would be frustrated by limiting the exemption to only individual vehicle owners who possess extraordinary knowledge and tools. For the exemptions to be more than empty promises, professional aftermarket companies must have the right to develop and distribute circumvention tools and replacement software to servicers; and servicers must have the right have the right to provide maintenance, repair, and customization services to business and individual consumers.

Auto Care believes that no violation of Section 1201(a) and (b) occurs where OEMs deploy technological measures to prevent aftermarket competition in products that use embedded software (rather than specifically to protect expressive non-functional aspects of copyrighted computer programs). However, Auto Care and its members cannot ignore the fact of chilling effects and the substantial costs and risks involved in litigating lawsuits that already have been or are being litigated on this precise question. Moreover, broad support was voiced in comments and during the roundtable sessions that Section 1201 was never intended to preclude otherwise lawful motor vehicle repair, regardless of whether it involves embedded control software. Therefore, Auto Care proposes on the following page, for discussion, a draft new exemption denominated as subsection Section 1201(l) which would exempt from liability under Section 1201(a) and (b) the provision of repair services for products that contain embedded control software, and the development and distribution of tools that make such services feasible. This discussion draft addresses the repair of products with embedded software generally. Auto Care would have no objection if this exemption were more limited in scope to only address the needs of businesses that compete in the motor vehicle aftermarkets for the development, manufacture, distribution or provision of parts and service, and the rights of the owners and lessees of such motor vehicles.

Auto Care thanks the Copyright Office for its continuing efforts in furtherance of this Study. We look forward to continuing our participation in this proceeding. Should the Office have any questions, please feel free to contact us at your convenience.

Respectfully submitted,

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Discussion Draft Proposal of Auto Care Association

Section 1201(l) PERMISSIBLE REPAIR OF PRODUCTS WITH EMBEDDED SOFTWARE.—

(1) CIRCUMVENTION PERMITTED. — Notwithstanding the provisions of subsection (a)(1)(A), it is not a violation of that subsection to circumvent a technological measure that effectively controls access to embedded software for the purpose of performing repair or service.

(2) PERMISSIBLE MANUFACTURE, IMPORTATION, AND TRAFFICKING. — Notwithstanding the provisions of subsection (a)(2) and subsection (b), it is not a violation of those subsections to manufacture, import, offer to the public, provide, or otherwise traffic in any technology, product, service, device, component, or part thereof that is designed and marketed for the purpose of, and that is capable of—

(A) circumventing a technological measure as permitted in paragraph (1), or

(B) circumventing a technological measure that effectively protects a right of a copyright owner under this title in embedded software for the purpose of performing repair or service.

(3) DEFINITIONS. — As used in this subsection, the following terms have the following meanings:

(A) "Embedded software" is a computer program or compilation protected under this title, which is embodied in a machine or product and which—

(i) controls the operation of such machine or product, or the interoperation of such machine or product with another machine or product, and,

(ii) is not designed or marketed for the primary purpose of causing the display, reproduction, or performance to the public of a work other than a computer program.

(B) "Repair or service" means—

(i) the replacement, repair, alteration, or customization of, or the provision of services for, a machine or product or the embedded software in such machine or product, by or on behalf of the owner or lessee of such machine or product, or

(ii) the making and distribution of physical parts or a computer program to perform such repair or service.

(C) "Machine or product" includes a part or component thereof, including a physical part or component and computer software.

Exhibit 1
Electronic Modules Controlled by Embedded Software